

VAJRAPU MANOJ KALYAN

+91 9346474431  bubbuvajrapu@gmail.com

PROFESSIONAL SUMMARY

Backend and AI-focused Computer Science graduate with internship experience in FastAPI, LangChain, RAG workflows, and biomedical AI systems. Skilled in developing scalable APIs, LLM-powered backend pipelines, and retrieval-augmented applications.

EDUCATION

Indian Institute of Information Technology Kalyani B.Tech - Computer Science and Engineering CGPA - 9.1	2021 - 2025 IIIT Kalyani, West Bengal
Ascent Junior College Class XII (AP BIE) - 96 %	2019 - 2021 Visakhapatnam, India
Indian Springs EM High School Class X (AP SSC) GPA - 10/10	2018 - 2019 Visakhapatnam, India

ACHIEVEMENTS

- **NPTEL:** Awarded *Elite Silver* certificate in **Java Programming**, also an *Elite Silver* in **C Programming**.
- **First Runner Up** Senior Category **Essay Competition** conducted by **TATA BUILDING INDIA** in 2018.
- **14th rank** in the **Competitive Test, SFRS Brilliant Student of Telugu States**, organized by The Society for Rural Scholars, Guntur in 2017-18.

EXPERIENCE

AI Backend Intern — Nuvo AI - Meril Life Sciences, Vapi, Gujarat (May 2025 – Nov 2025)

- Developed backend pipelines for biomedical retrieval and AI-powered report generation using FastAPI, LangChain, and vector retrieval workflows.
- Integrated Neo4j-based knowledge graph retrieval to improve contextual relevance in biomedical QA systems.
- Integrated and deployed LLM workflows using Gemini, Ollama, and LLaMA models for biomedical AI applications.
- Developed prompt orchestration and context retrieval workflows to improve biomedical response relevance and consistency.
- Collaborated with AI and backend teams to integrate LLM-powered workflows into biomedical research applications.

PROJECTS

Choose Your Own Adventure AI

January 2025 - February 2025

Full-Stack Interactive Storytelling Application

Built a full-stack AI storytelling platform using FastAPI and React with asynchronous GPT-based story generation, SQLite persistence, and job status tracking.

- Implemented dynamic branching story logic with AI-generated responses based on user decisions.
- Developed asynchronous backend workflows to support responsive real-time story generation.
- Tech Stack: Python, FastAPI, React, SQLite, OpenAI GPT APIs

Agricultural Yield Prediction Based on Weather Patterns

July 2024 - November 2024

- Developed a crop yield prediction model using weather and agricultural datasets.
- Performed feature engineering using ERA5 climate data to improve prediction accuracy.
- Trained and evaluated Random Forest and XGBoost models for yield estimation.
- Tech Stack: Python, Scikit-learn, Seaborn, Matplotlib, NumPy

Hermitian Code-Based McEliece Cryptosystem Project

Jan 2024 - April 2024

- Designed and implemented a post-quantum cryptographic system based on the McEliece framework with Hermitian codes.
- Developed key generation, encryption, and decryption workflows with security-focused evaluation.
- Tech Stack: Python, PyCryptodome, Galois Field, Git, VS Code.

SKILLS

Languages: Python, C++, JavaScript, C

Backend & APIs: FastAPI, REST APIs, Async Programming, Pydantic

AI & Data: LangChain, RAG pipelines, Prompt Engineering, Ollama, Scikit-learn

Databases: MySQL, MongoDB, PostgreSQL, Neo4j

Tools & Platforms: Git, GitHub, Docker, Linux

Core CS: Data Structures and Algorithms, DBMS, OOPs

POSITIONS OF RESPONSIBILITY

- Held the position of Secretary for IIIT Kalyani's Symphony Club, managing event organization and engaging 50+ participants.
- Led the Garnet Clubhouse for 2 consecutive years, managing 200+ students and recognized as "Best Club Leader" among all school clubs.